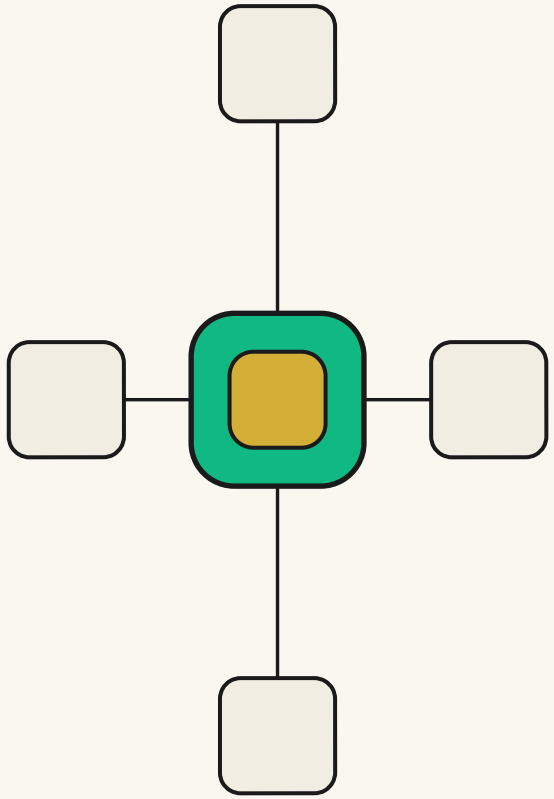
RepoScope

RepoScope

One code graph. Four ways in: the canvas, the CLI, an MCP server, and your editor.

`v2.0 · source-available · github.com/adityapatel5912/RepoScope`



Every repository is a graph. We still read it as a pile of files.

1. REPEATED CONTEXT TAX

Agents re-read the file tree, pay for the same context repeatedly, and diverge from human answers.

2. INVISIBLE BLAST RADIUS

Humans grep across files and guess at downstream impact before making a change.

3. TRIBAL ONBOARDING

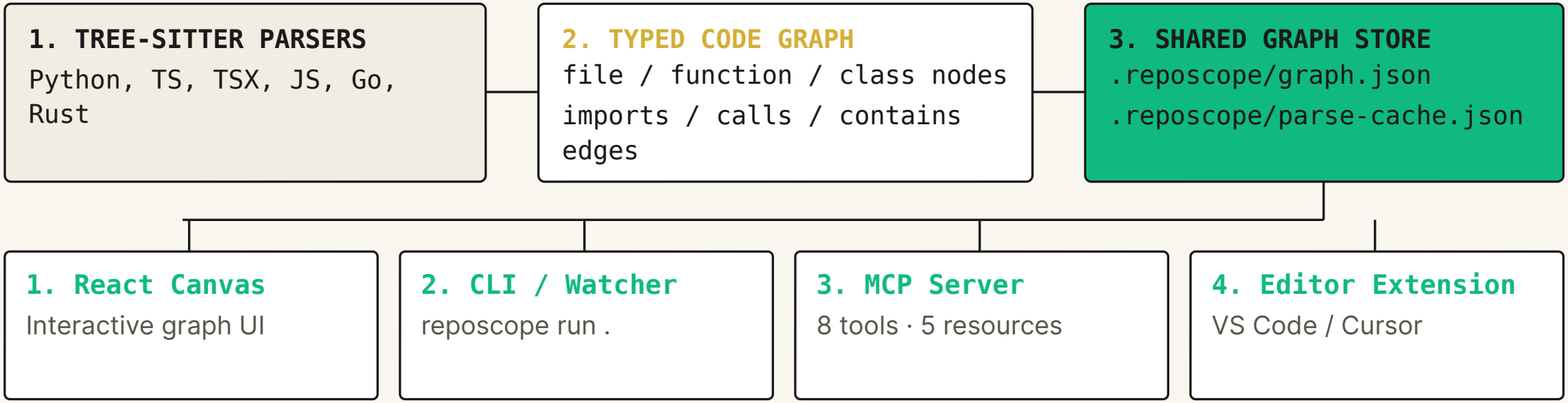
Onboarding is slow because system structure lives in scattered tribal knowledge.

AUDIENCE · DEVELOPERS JOINING A CODEBASE · REVIEWERS · MAINTAINERS · AGENT TEAMS

No shared source of truth exists between people and coding agents inspecting the same repository.

One index. Four ways in.

Every surface reads .reposcope/graph.json, so humans and agents quote the same answer.



STATIC LIMIT · CALL RESOLUTION IS STATIC

Dynamic dispatch, reflection, and generated code can be missed by static AST analysis.

Read the shape of the repo before reading the files.

SMALL vs. LARGE REPO SCALING

- Small repos: connectivity pyramid (root → backbone → core → services → UI → config → docs).
- Large repos: top-level module cards + module drawer.

FOUR VIEWS & MODULE DRAWER

1. Grouped by module 2. Architecture clusters
3. Flat inspection 4. Git-churn overlay

- Drawer: symbols, importers, deps, PNG/SVG export.

[REPO SCREENSHOTS · docs/screenshots/readme-hero.png]

5 ARCHITECTURE CLUSTERS

Client/UI · Backend/API · Storage/Data · External/Integrations · Compute/Workers

+ readme-graph-grouped · clusters · flat · churn · module-drawer

LIMIT · RENDERING COLLAPSE

Grouped view is a rendering collapse, not a second graph. WebGL for very large graphs is planned.

From repository to file to symbol without losing the thread.

[SCREENSHOT · readme-side-panels.png]

LEFT PANEL · GITINGEST FILE TREE

Search filter, hierarchical repo tree copy, and single-file download.

RIGHT PANEL · SYMBOL & EDGE INSPECTOR

Incoming/outgoing edges, callers, dependencies, and IDE deep links.

1. FILE & SYMBOL LINEAGE

Inspect functions, classes, importers, and cross-file call edges in place.

2. EXPORT & QUICK ACTIONS

Export graph as PNG or SVG, generate repo trees, and open IDE deep links.

3. ONE-CLICK DEMO REPOS

Load pre-configured demo repositories for an immediate starting point.

The graph answers questions grep cannot.

1. PAGERANK & HUBS
Ranks central modules & symbols.

Local: 348 files, 1,863 nodes, 3,419 edges.

2. SCC CYCLES
Finds circular import & symbol loops via Strongly Connected Components.

3. DEAD CODE
Static reachability flags unreachable functions & classes.

4. CHURN & RULES
Git-churn hotspots
+ .reposcope.yml per-import boundary checks.

5. CRITICAL PATH
Critical-path analysis API endpoint shipped (UI panel planned).

6. BUDGETED REPO MAP
PageRank-ordered map cuts agent context 98.5%–99.8%.

LIMIT · STATIC ANALYSIS, NOT RUNTIME TRACING

All centrality, cycle, reachability, and rule checks operate on static Tree-sitter ASTs, not live execution traces.

Ask the repository. Get an answer you can check.

[SCREENSHOT · docs/screenshots/chat.png]

Label: “earlier UI generation”

CITED STREAMING RESPONSE

- SSE-streamed answers from graph context
- Clickable file, function & line citations
- Markdown, tables, code blocks & copy buttons
- Actionable Next steps section
- Browser voice input where supported

PROVIDER CHAIN & 429 MULTI-KEY ROTATION

OpenRouter (primary) → Groq (secondary) → NVIDIA NIM (fallback), with automatic multi-key rotation on HTTP 429 rate limits.

IN-MEMORY BYOK GITHUB TOKEN

Private repo tokens stay in memory only (readme-byok.png).

REQUIREMENT · PROVIDER API KEY

Chat requires >= 1 LLM key; the graph and local mode do not.

See the blast radius before the pull request.

1. DIRECT & TRANSITIVE DEPENDENTS

Reverse BFS highlights affected callers on the canvas and computes risk score, narrative, and suggested tests.

2. PR BOT & MERMAID IMPACT PREVIEW

Generates a GitHub-ready review comment with a Mermaid impact graph plus an included GitHub Action workflow.

[REPO SCREENSHOTS · docs/screenshots/readme-impact.png & docs/screenshots/readme-prbot.png]

VERIFICATION BOUNDARY · LIVE PR COMMENT NOT VERIFIED IN FINAL PASS

Posting a live PR comment requires a real GitHub pull request and pull-requests: write permission; only the preview path was verified.

Turn the graph into a path through the codebase.

1. STUDENT MODE (3 LEVELS)

- Level 1: Entry points → Level 2: Core modules → Level 3: Utilities
- Per-file “why it matters” notes & checkpoint quizzes
- Good First Issues mined from low-traffic files

2. INTERACTIVE CODE TOURS

- Numbered architectural stops with smooth camera pans
- Node and edge highlights at each stop
- Optional voice narration via OpenRouter TTS or browser speech

[REPO SCREENSHOTS · docs/screenshots/readme-student.png & docs/screenshots/readme-tour.png]

DEPENDENCY SPLIT · GRAPH RANKING vs. LLM GENERATION

Structural ordering and low-traffic file mining come from the graph; generated explanations and quizzes require an LLM key.

Your repository does not have to leave your machine.

VERIFIED LOCAL MODE · ZERO UPLOAD · NO GITHUB ACCOUNT, TOKEN, OR CLONE REQUIRED

reposcope run . indexes in place (348 files, 1,863 nodes, 3,419 edges), serves 127.0.0.1:4100, and streams /api/local/events SSE edits.

LOCAL CLI & SESSION RESUME

- `reposcope index .` | `--format sqlite`
- `reposcope mcp .` | `resumes after restart`

WASM BROWSER PREVIEW (75 / 315 / 310)

- showDirectoryPicker + Tree-sitter WASM
- 75 files, 315 nodes, 310 edges in Chrome/Edge

BROWSER PREVIEW BOUNDARY · CLIENT-ONLY WASM (NO CROSS-FILE SYMBOL RESOLUTION)

Clearly labelled; hides server-backed chat/tracking/Student/map/security/tour/impact. Cancel keeps prior preview; actionable errors on denied/empty.

The same graph, available to Claude, Cursor, OpenCode, and any MCP client.

8 MCP TOOLS (STDIO & STREAMABLE HTTP; NO SSE)

- get_architecture_overview
- calculate_blast_radius
- get_dependency_lineage
- explain_architectural_rule
- get_repo_map
- find_dependency_cycles
- get_hotspots
- find_unreachable_symbols

Verified: 8 tools, 5 resources, 3 read rounds, clean ping.

5 MCP RESOURCES

- reposcope://graph/summary
- reposcope://hotspots
- reposcope://repo-map
- reposcope://dead-code
- reposcope://cycles

```
opencode mcp list → reposcope connected
```

KNOWN EDGE · ONE OPENCODE RESOURCE-READ BRIDGE STALLED

One OpenCode resource-read bridge stalled during testing; direct stdio resource reads and all 8 tool calls work cleanly.

The graph follows the file you are editing.

1. ACTIVITY BAR & STATUS BAR

Focus-first mini-graph around open file.

Verified status bar: “RepoScope: 12 dependents | Risk: Critical”.

2. CODELENS & EXPLORER MENU

Verified CodeLens: “View Callers (6)” above exported symbols.

Explorer: “RepoScope: Focus this node in the canvas”.

3. URI HANDLER & OFFLINE RECOVERY

reposcope://open opens backend/main.py:392.

Offline state displays “RepoScope: offline” (never fake zero).

4. THIN HTTP CLIENT (NEVER PARSES CODE)

Reads backend over HTTP (:8000 default, :4100 local mode).

Open VSX v2.0.1: 288 downloads, searchable.

DISTRIBUTION & MANUAL CHECKS · OPEN VSX verified:false · MARKETPLACE PAT · CURSOR/VSCODIUM

Open VSX namespace is still verified:false; VS Code Marketplace is blocked on an Azure DevOps PAT; Cursor/VSCodium were not exercised in final pass.

More than a canvas.

1. TRACKING (tracking.png)

- Commits, PRs, issues, releases & breaking-change flags
- On-demand GitHub polling stored in data/repo_state.json
- Subject to GitHub API rate limits

2. SECURITY — HEURISTIC (security.png)

- Redacted secrets, leaked .env detection, signature breaks
- Vulnerable dependency pins, taint endpoint & remediation
- Heuristic checks—not a replacement for a dedicated scanner

3. REVERSE BUILD & SCAFFOLD (scaffold.png)

- Turns a repo into a “Build me a…” Markdown prompt
- Stage 1 / 2 / 3 starter, ZIP download & StackBlitz launch
- Never downloads the whole codebase

4. WORKSPACE TOOLS

- GitIngest file tree, search filter & single-file download
- High-resolution PNG & SVG graph exports
- One-click demo repositories & node quick actions



One graph. Real evidence. Known edges.

CTA: github.com/adityapatel5912/RepoScope · run reposcope run .

VERIFIED EVIDENCE (v2.0)	4 MANUAL CHECKS & GAPS	SOURCE-AVAILABLE LICENSE 1.0
• 335 backend + 20 ext tests pass • TS clean; build: index-CMSb0cnC.js • Local: 348 files, 1,863 nodes, 3,419 edges • Browser WASM: 75 files, 315 nodes, 310 edges • MCP: 8 tools, 5 resources, 3 read rounds • /api/health: 200 (2.0.0); Open VSX: 288 dl	• Manual: (1) Cursor/VSCodium, (2) Open VSX badge (verified:false), (3) Render 2.0.0 redeploy (now 1.0.0), (4) Marketplace PAT • Gaps: WebGL, kuzu, monorepo edges, ownership, critical-path UI, visual baselines, 50k test, fixtures, live PR & LLM	• RepoScope Source-Available License 1.0: use and modify in your own projects; no republishing or commercial exploitation without written permission. • Not OSI Open Source. • Earlier MIT releases (incl. ext 2.0.1) keep MIT grants.